

MULTI-SEGMENTAL ROTATION



Test Objective for the Multi-Segmental Rotation Assessment:

The Multi-Segmental Rotation assessment tests for normal rotational mobility trunk, pelvis, hips, knees and feet.

How to Perform the Multi-Segmental Rotation Assessment:

The patient assumes the starting position by standing erect with feet together, toes pointing forward and arms relaxed to the sides. The patient then rotates the entire body—hips, shoulders and head—as far as possible to the right while the foot position remains unchanged. Have the patient return to the starting position, and rotate to the left.

Functional Criteria:

- 100 degrees of total rotation (50 degrees from pelvis, additional 50 degrees from torso)
- Maintains posture
- Maintains foot position
- Without excessive effort and/or motor control

Common Compensations:

- Hip and/or knee flexion
- Spine and/or pelvis deviation
- Protraction/retraction of shoulder girdle
- Loss of foot/ankle position

Tips for Testing:

- Observe from the back.
- Foot position should remain unchanged throughout the movement.
- Do not coach the movement; simply repeat the instructions if needed.
- Was there pain?

Additional Information:

Since both sides are tested simultaneously with the feet together, the externally rotating hip is also extending, and this can limit rotation. Pay close attention to each segment of the body, the hips, trunk and head. One area may be hypermobile due to restrictions in an adjacent segment.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Multi-Segmental Rotation Breakout Rationale

Multi-segmental rotation dysfunctions can be due to trunk/spine rotation dysfunction, hip internal or external rotation dysfunction or even problems rotating below the knee, or a combination of all these areas. Regardless, we start the Multi-Segmental Rotation Breakout with the trunk/spine and follow the instructions from there.

Spine Rotation Problem

The first test, the Seated Rotation Test, removes all non-trunk related possible problems, such as the hips and tibias. The test isolates trunk rotation. If the Seated Rotation Test is functional and non-painful bilaterally, assume the trunk/spine is clear and proceed to the Lower Quarter Rotation Flowcharts. If the test is dysfunctional and/or painful, we need to change the stability requirements and check the parts of the trunk - thorax and lumbar regions.

Thorax Extension/Rotation Problem

If the Seated Rotation Test is dysfunctional and/or painful, we must further breakout spine rotation. To differentiate between the thorax and lumbar regions, we use the Active Lumbar Locked (IR) Extension/Rotation Test. The lumbar lock position allows better isolation of the thorax and reduces the contribution from the lumbar spine. Placing the arm behind the back in an internally rotated position allows a reduction in scapular involvement and minimizes anterior shoulder and chest dysfunction.

If thorax mobility is functional and non-painful bilaterally, then assume the thorax is meeting minimum active movement mobility and motor control - continue to evaluate the lumbar region with the Prone on Elbow Test. If there is dysfunction and/or pain (either bilateral or unilateral) with Active Lumbar Locked Extension/Rotation Test, we retest the motion passively to differentiate between a mobility or stability/motor control dysfunction.

If the Passive Lumbar Locked (IR) Extension/Rotation motion is dysfunctional, with or without pain, there is a thorax extension/rotation mobility dysfunction. At this point, we have ruled in a thorax mobility dysfunction - we have not ruled out lumbar dysfunction. A clinical pearl within the SFMA is that a Thorax Mobility Dysfunction finding is a priority over a Lumbar Dysfunction finding based on our mobility/stability model. Therefore, we suggest treating the thorax mobility dysfunction prior to assessing the lumbar region.

If the Passive Lumbar Locked Extension/Rotation test is now functional, with or without pain, there is a thorax extension/rotation SMCD. If mobility of the thorax is cleared bilaterally passively, with or without pain, we must continue to assess the lumbar spine by proceeding to the Prone on Elbow Test.

Lumbar Extension/Rotation Problem

To assess the lumbar region, we use the Prone on Elbow Test. In this position, the thorax is already extended, so the stress will now be placed on the lumbar region. If extension from this position creates pain, stop and note where the pain is and come back and examine the region after completing the entire SFMA.

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If lumbar extension/rotation is functional and non-painful bilaterally, we must first go back and see what happened with the thorax. If the thorax was an SMCD (with or without pain), then we can assume the lumbar is normal. If the thorax was also functional, then we must assume there is a weight-bearing spine extension SMCD.

If there is dysfunction and/or pain with lumbar extension/rotation, we must proceed to passive testing to differentiate between mobility or stability/motor control dysfunction. If the Passive Prone on Elbow Test is functional and non-painful, assume the patient has a weight bearing spine extension SMCD.

If the Passive Prone on Elbow Test is dysfunctional and non-painful, then assume the patient has a lumbar extension/rotation mobility dysfunction.

If the Passive Prone on Elbow Test creates pain, stop and note where the pain is and come back and examine the region after completing the entire SFMA.

Once the spine has been assessed, we need to look at the lower body for rotation. The lower quarter assessment includes both hip rotation and tibial rotation. First, we will look at lower quarter external rotation.

Lower Quarter External Rotation Problem

We will begin this flowchart by reducing the postural stabilization requirements for active external hip rotation by placing the patient in a prone position. Since the hip is extended in our top tier test, we will assess hip rotation with the hip extended in the Active Prone External Hip Rotation Test. If prone external hip rotation is functional and non-painful, consider hip extended external rotation normal and proceed to the Active Seated External Tibial Rotation Test.

If the Active Prone External Hip Rotation Test is dysfunctional and/or painful, then we proceed to the Stabilized Prone External Hip Rotation Test. If by activating the core, essentially re-positioning the pelvic position, creates functional and non-painful external hip rotation, the problem is a core/pelvic orientation SMCD. If the stabilization does not change anything and the test is still dysfunctional and/or painful, proceed to the Passive Prone External Hip Rotation Test.

If the Passive Prone External Hip Rotation Test is now functional and non-painful, there is a weight-bearing or active external hip rotation stability/motor control dysfunction. If this test is dysfunctional, it indicates an external hip rotation mobility dysfunction.

If there is pain with the Passive Prone External Hip Rotation Test, note where the pain is and come back and examine the region after completing the entire SFMA. No matter your finding for external hip rotation, we must proceed to assess tibial external rotation.

If the Active Seated External Tibial Rotation Test is functional and non-painful, consider external tibial rotation normal. If this test is dysfunctional and/or painful, proceed to the Passive External Tibial Rotation Test to differentiate between a mobility and stability/motor control dysfunction.

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If the passive test is functional and non-painful, there is an external tibial rotation SMCD. If the Passive External Tibial Rotation Test is dysfunctional, there is an external tibial rotation mobility dysfunction. If there is pain with passive testing, note where the pain is and come back and examine the region after completing the entire SFMA.

Now we must check internal hip rotation and internal tibial rotation.

Lower Quarter Internal Rotation Problem

We will begin this flowchart by reducing the postural stabilization requirements for active internal hip rotation by placing the patient in a prone position. Since the hip is extended in our top tier test, we will assess hip rotation with the hip extended in the Active Prone Internal Hip Rotation Test. If prone internal hip rotation is functional and non-painful, consider hip extended internal rotation normal and proceed to the Active Seated Internal Tibial Rotation Test.

If the Active Prone Internal Hip Rotation Test is dysfunctional and/or painful, then we proceed to the Stabilized Prone Internal Hip Rotation Test. If by activating the core, essentially re-positioning the pelvic position, creates functional and non-painful internal hip rotation, the problem is a core/pelvic orientation SMCD. If the stabilization does not change anything and the test is still dysfunctional and/or painful, proceed to the Passive Prone Internal Hip Rotation Test.

If the Passive Prone Internal Hip Rotation Test is now functional and non-painful, there is a weight-bearing or active internal hip rotation stability/motor control dysfunction. If this test is dysfunctional, it indicates an internal hip rotation mobility dysfunction.

If there is pain with the Passive Prone Internal Hip Rotation Test, note where the pain is and come back and examine the region after completing the entire SFMA. No matter your finding for internal hip rotation, we must proceed to assess internal tibial rotation.

If the Active Seated Internal Tibial Rotation Test is functional and non-painful, consider tibial internal rotation normal. If this test is dysfunctional and/or painful, proceed to the Passive Internal Tibial Rotation Test to differentiate between a mobility and stability/motor control dysfunction.

If the passive test is functional and non-painful, there is an internal tibial rotation SMCD. If the Passive Internal Tibial Rotation Test is dysfunctional, there is an internal tibial rotation mobility dysfunction. If there is pain with passive testing, note where the pain is and come back and examine the region after completing the entire SFMA.

This completes the Multi-Segmental Rotation Breakout.

MULTI-SEGMENTAL ROTATION BREAKOUTS



Selective Breakout Tests for Multi-Segmental Rotation:

Spine



Seated
Rotation Test



Active Lumbar
Locked Extension/
Rotation Test



Passive Lumbar
Locked Extension/
Rotation Test



Active Prone on
Elbow Extension/
Rotation Test



Passive Prone on
Elbow Extension/
Rotation Test

Lower Quarter External Rotation



Active Prone
External Hip
Rotation Test



Stabilized Prone
External Hip
Rotation Test



Passive Prone
External Hip
Rotation Test



Active Seated
External Tibial
Rotation Test



Passive Seated
External Tibial
Rotation Test

Lower Quarter Internal Rotation



Active Prone
Internal Hip
Rotation Test



Stabilized Prone
Internal Hip
Rotation Test



Passive Prone
Internal Hip
Rotation Test

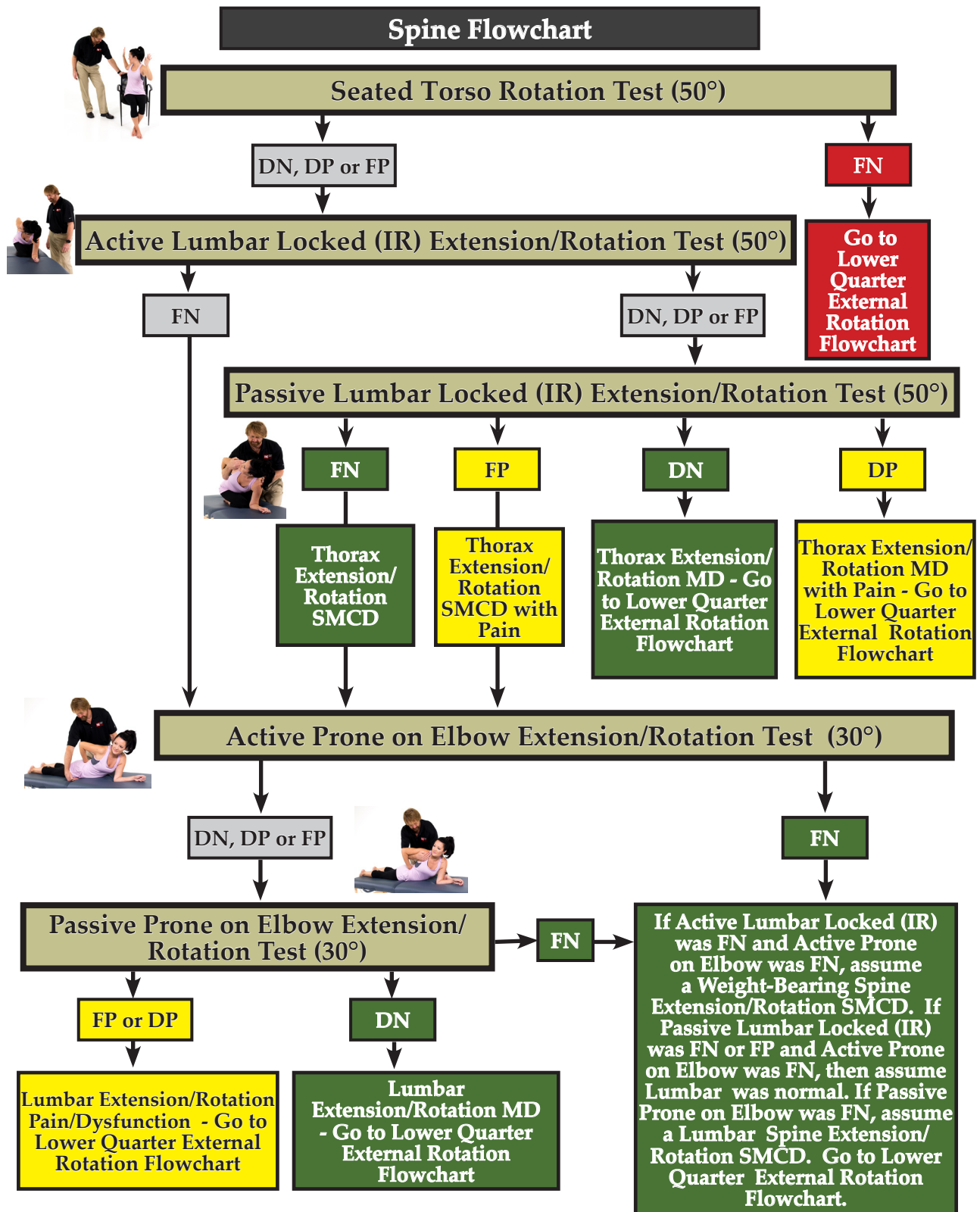


Active Seated
Internal Tibial
Rotation Test



Passive Seated
Internal Tibial
Rotation Test

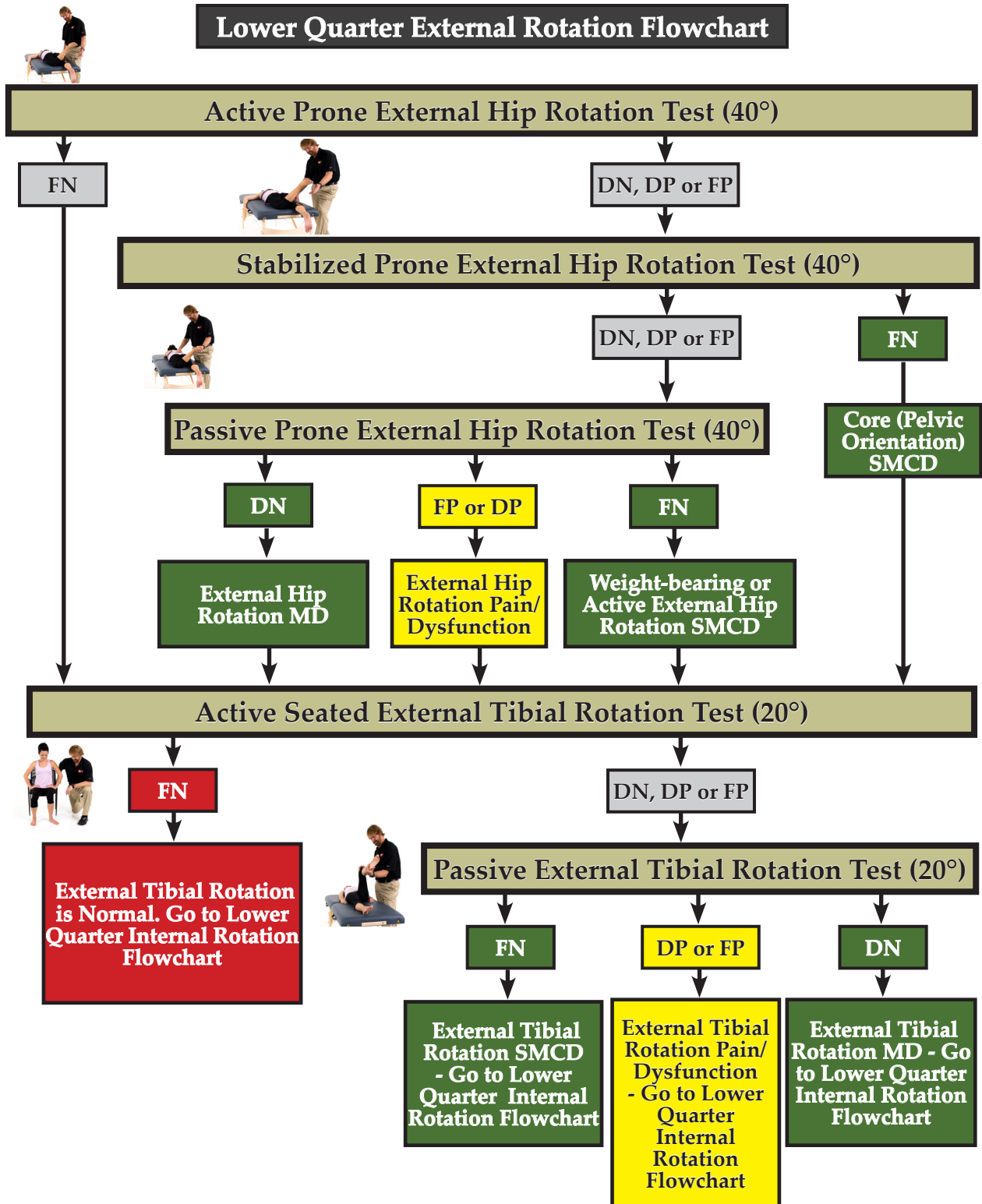
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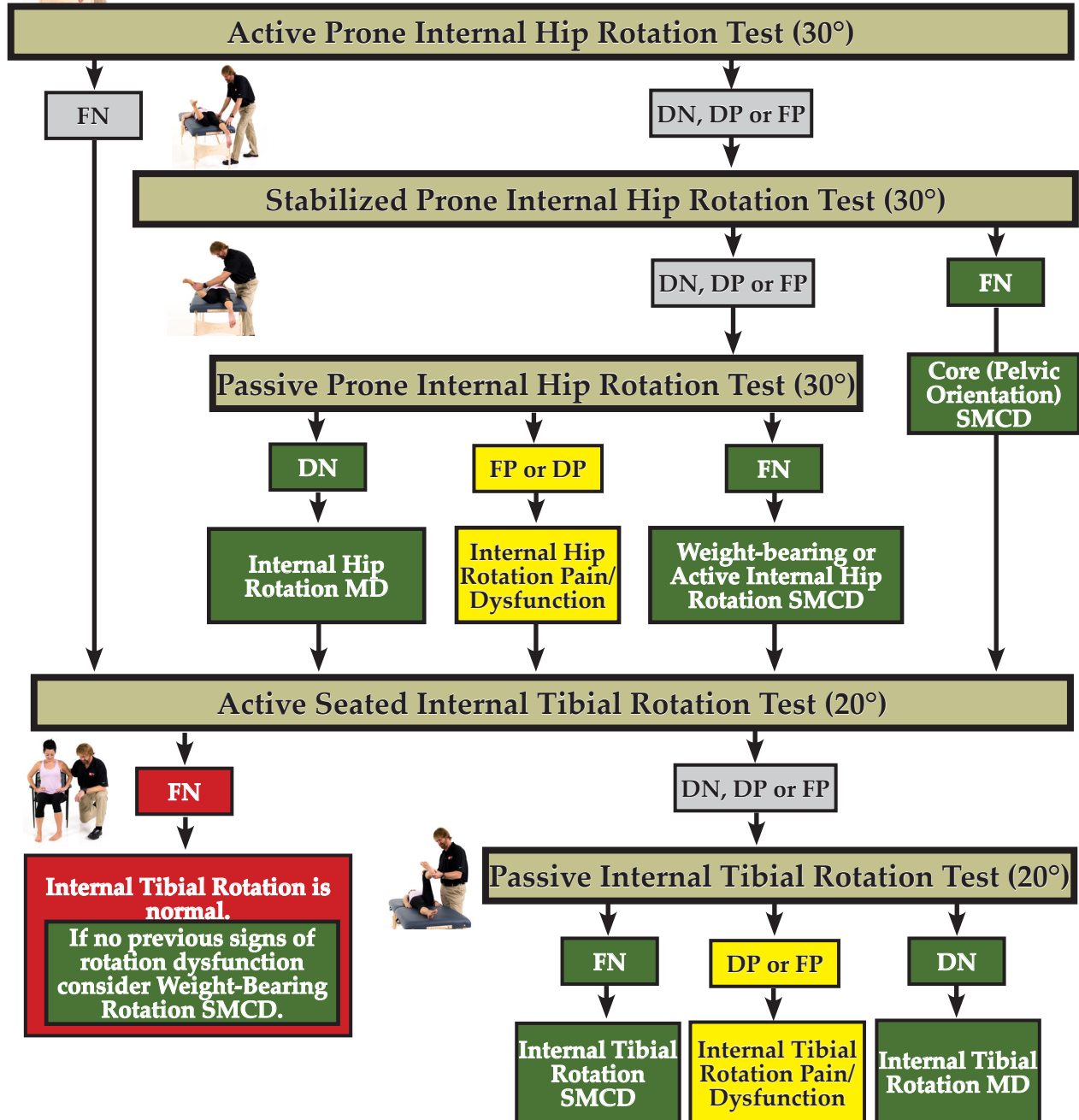
Lower Quarter External Rotation Flowchart



MULTI-SEGMENTAL ROTATION BREAKOUTS



Lower Quarter Internal Rotation Flowchart



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Seated Rotation Test:

To test for trunk rotation bilaterally, removing the lower body (hips, knees, and ankles).

How to Perform the Seated Rotation Test:

Instruct the patient to get into a seated position with the knees and feet together, sitting tall so their body is in an upright and erect posture. Place a dowel across their shoulders behind their neck and have them place their hands under the dowel on both sides. Have the patient cross their right leg in front of left, placing their right foot flat on the ground. Once in position, rotate the torso to the right as far as possible. Repeat the process for left rotation - left leg crossing in front of right leg and then torso rotation left.

Functional Criteria:

Normal range for trunk rotation is 50 degrees bilaterally.

Additional Information

The dowel is used to limit scapular recruitment. If you do not have a dowel, arm position should be 90 degrees shoulder abduction and 90 degrees elbow flexion.

Crossing the legs reduces involvement from the pelvis - make sure to monitor for any pelvic rotation.

Possible Findings:

- FN: 50 degrees or greater of trunk rotation bilaterally, no pain
 - » Consider the trunk clear for extension/rotation. Proceed to the Lower Quarter External Rotation Flowchart.
- FP: 50 degrees or greater of trunk rotation bilaterally, pain
 - » Proceed to the Active Lumbar Locked (IR) Extension/Rotation Test.
- DN: Less than 50 degrees of trunk rotation unilaterally or bilaterally, no pain
 - » Proceed to the Active Lumbar Locked (IR) Extension/Rotation Test.
- DP: Less than 50 degrees of trunk rotation unilaterally or bilaterally, pain
 - » Proceed to the Active Lumbar Locked (IR) Extension/Rotation Test.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Active Lumbar Locked (IR) Extension/Rotation Test:

To test the combined active extension and rotation in the thorax in an unloaded posture.

How to Perform the Active Lumbar Locked (IR) Extension/Rotation Test:

Instruct the patient to get into a full prone rocking position (butt to heels) with their forearms together in front of their knees. Their knees do not need to be together. Place their right hand and forearm behind on the small of their back with their palm facing upwards. To set their scapula in proper position, the shoulder should be in a mid-axial position. Their left forearm and hand remain on the testing surface in the start position. Have the person rotate their head and thorax to the right as far as possible while maintaining the set-up position. Compare both sides.

Functional Criteria:

When using a line through the AC Joints as the reference line, normal thorax extension/rotation is 50 degrees relative to the horizontal bilaterally, without lateral flexion/side bending.

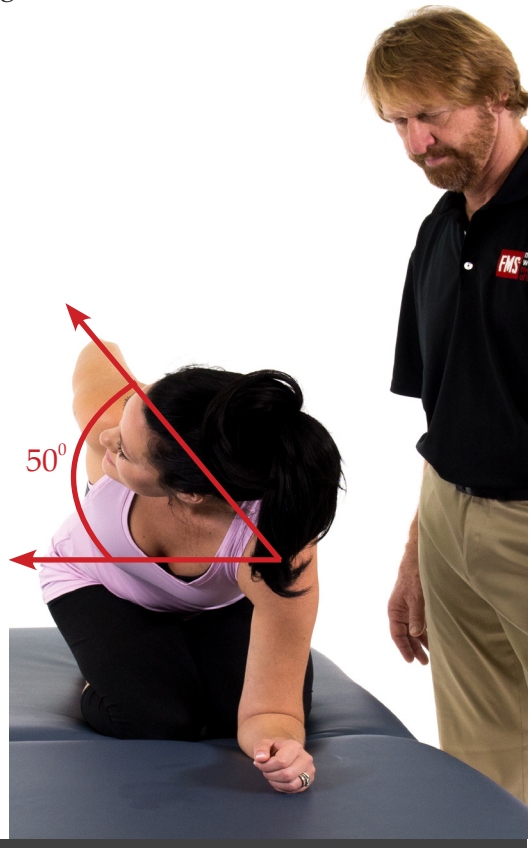
Additional Information:

The prone rocking position helps eliminate the lumbar spine from contributing to the extension/rotation. This can also differentiate between right and left thorax restrictions.

If the patient has pain with the shoulder IR test position, it is appropriate to use a chest position for the arm - their right arm across their chest, resting right hand on their left shoulder.

Possible Findings:

- FN: 50 degrees or greater of active thorax extension/rotation, no pain
 - » Consider the thorax extension/rotation normal. Proceed to the Active Prone on Elbow Extension/Rotation Test.
- FP: 50 degrees or greater of active thorax extension/rotation, pain
 - » Proceed to Passive Lumbar Locked (IR) Extension/Rotation Test.
- DN: Less than 50 degrees of active thorax extension/rotation on one side (unilateral DN), or on both sides (bilateral DN) no pain
 - » Proceed to Passive Lumbar Locked (IR) Extension/Rotation Test.
- DP: Less than 50 degrees of active thorax extension/rotation on one side (unilateral DP), or on both sides (bilateral DP), pain
 - » Proceed to Passive Lumbar Locked (IR) Extension/Rotation Test.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Passive Lumbar Locked (IR) Extension/Rotation Test:

To test the combined passive extension and rotation of the thorax in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Lumbar Locked (IR) Extension/Rotation Test:

Instruct the patient to get into a full prone rocking position (butt to heels) with their forearms together in front of their knees. Place their right hand and forearm behind on the small of their back with their palm facing upwards. To set their scapula in proper position, the shoulder should be in a mid-axial position. Their left forearm and hand remain on the testing surface in the start position. Stabilize on the outside of their left shoulder with your left hand and at their hip, by placing your right hip against their left hip. This allows you to create a tunnel for rotation. Thread your right arm through their elbow, placing your right forearm tightly along their trunk with ulnar border of your hand over their coracoid process. Passively rotate the thorax to the right as far as possible while maintaining the set-up position. Compare both sides.

Functional Criteria:

When using a line through the AC Joints as the reference line, normal thorax extension/rotation is 50 degrees relative to the horizontal bilaterally, without lateral flexion/side bending.

Additional Information:

The prone rocking position helps eliminate the lumbar spine from contributing to the extension/rotation. This can also differentiate between right and left upper thorax restrictions.

If the patient has pain with the shoulder IR test position, it is appropriate to use a chest position for the arm - their right arm across their chest, resting right hand on their left shoulders.

Possible Findings:

- FN: 50 degrees or greater of passive thorax extension/rotation, no pain
 - » Indicates a thorax extension/rotation stability/motor control dysfunction. Proceed to the Active Prone on Elbow (IR) Extension/Rotation Test.
- FP: 50 degrees or greater of passive thorax extension/rotation, pain
 - » Indicates pain in the thorax - but this could be due to Lumbar mobility problem. Proceed to the Active Prone on Elbow Test.
- DN: Less than 50 degrees of passive thorax extension/rotation on one side (unilateral DN) or both sides (bilateral DN), no pain
 - » Indicates a unilateral/bilateral thorax extension/rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Lower Quarter External Rotation Flowchart.
- DP: Less than 50 degrees of passive thorax extension/rotation on one side (unilateral DP) or both sides (bilateral DP), pain
 - » Indicates a painful unilateral/bilateral thorax extension/rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Lower Quarter External Rotation Flowchart.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Active Prone on Elbow (IR) Extension/Rotation Test:

To test the active extension and rotation of the lumbar region in an unloaded posture.

How to Perform the Active Prone on Elbow Unilateral Ext./Rot. (IR) Test:

Instruct the patient to lie prone, then push up to elbows and position forearms together. Place their right hand behind the back with their palm facing upward. The left forearm and hand remain in the start position. Have the person rotate the thorax to the right as far as possible while maintaining the set-up position. Compare both sides.

Functional Criteria:

When using a line through the AC Joints as the reference line, normal lumbar extension/rotation is 30 degrees relative to the floor bilaterally. The hips should remain in contact with the floor during movement.

Additional Information:

Both the thorax and lumbar contribute to the 30 degrees of movement seen from the AC Joints - 20 degrees coming from the thorax and 10 degrees from the lumbar region.

If the patient has pain with the shoulder IR test position, it is appropriate to use a chest position for the arm - their right arm across their chest, resting right hand on their left shoulder.

Possible Findings:

- FN: 30 degrees or greater of extension/rotation, no pain
 - » If the Active Lumbar Locked (IR) was FN, consider this a weight-bearing spine extension stability/motor control dysfunction. If Passive Lumbar Locked (IR) was FN/FP, consider lumbar normal. Proceed to the Lower Quarter External Rotation Flowchart.
- FP: 30 degrees or greater of extension/rotation, pain
 - » Proceed to the Passive Prone on Elbow Extension/Rotation Test.
- DN: Less than 30 degrees of extension/rotation on one side (unilateral DN) or on both sides (bilateral DN), no pain
 - » Proceed to the Passive Prone on Elbow Extension/Rotation Test.
- DP: Less than 30 degrees of extension/rotation on one side (unilateral DN) or on both sides (bilateral DN), pain
 - » Proceed to the Passive Prone on Elbow Extension/Rotation Test.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Passive Prone on Elbow (IR) Extension/Rotation Test:

To test the passive extension and rotation of the lumbar region in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone on Elbow (IR) Extension/Rotation Test:

Instruct the patient to lie prone, then push up to elbows and position forearms together. Place their right hand behind the back with their palm facing upward. The left forearm and hand remain in the start position. Stabilize on the outside of their left shoulder with your left hand and at their hip, by placing your right hip against their left hip. This allows you to create a tunnel for rotation. Thread your right arm through their elbow, placing your right forearm tightly along their trunk with ulnar border of your hand over their coracoid process. Passively rotate the thorax to the right as far as possible while maintaining the set-up position. Compare both sides.

Functional Criteria:

When using a line through the AC Joints as the reference line, normal lumbar extension/rotation is 30 degrees relative to the floor bilaterally. The hips should remain in contact with the floor during movement.

Additional Information:

Both the thorax and lumbar contribute to the 30 degrees of movement seen from the AC Joints - 20 degrees coming from the thorax and 10 degrees from the lumbar spine.

If the patient has pain with the shoulder IR test position, it is appropriate to use a chest position for the arm - their right arm across their chest, resting right hand on their left shoulder.

Possible Findings:

- FN: 30 degrees or greater of extension/rotation, no pain
 - » If the finding is FN on both sides, there is a lumbar extension/rotation stability/motor control dysfunction. Proceed to the Lower Quarter Rotation Flowchart.
- FP: 30 degrees or greater of extension/rotation, pain
 - » Indicates pain in the lumbar region - complete local biomechanical exam. Proceed to the Lower Quarter Rotation Flowchart.
- DN: Less than 30 degrees of extension/rotation on one side (unilateral DN) or on both sides (bilateral DN), no pain
 - » Indicates a unilateral/bilateral lumbar extension/rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Lower Quarter Rotation Flowchart.
- DP: Less than 30 degrees of extension/rotation on one side (unilateral DN) or on both sides (bilateral DN), pain
 - » Indicates a painful unilateral/bilateral lumbar extension/rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Lower Quarter Rotation Flowchart.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Active Prone External Hip Rotation Test:

To test active external hip rotation in an unloaded posture with the hip extended.

How to Perform the Active Prone External Hip Rotation Test:

Instruct the patient to lie face down on the table/floor with the knee flexed. Have the patient externally rotate their hip by allowing their lower leg to fall towards midline while keeping the knee flexed. Have the patient externally rotate the hip as far as possible. Repeat the test on the other side.

Functional Criteria:

Normal range for external hip rotation is 40 degrees bilaterally.

Additional Information:

Monitor for pelvic rotation.

Possible Findings:

- FN: 40 degrees or greater of active external hip rotation, no pain
 - » Consider hip external rotation normal in the extended position. Proceed to the Active Seated External Tibial Rotation test.
- FP: 40 degrees or greater of active external hip rotation, pain
 - » Proceed to the Stabilized Prone External Hip Rotation Test.
- DN: Less than 40 degrees of external hip rotation, no pain
 - » Proceed to the Stabilized Prone External Hip Rotation Test.
- DP: Less than 40 degrees of external hip rotation, pain
 - » Proceed to the Stabilized Prone External Hip Rotation Test.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Stabilized Prone External Hip Rotation Test:

To test if there is an underlying SMCD of the core that may be limiting hip mobility. Pelvic orientation can directly affect the spacing in the acetabulum and therefore give the impression that there is a hip joint mobility problem, when in reality it may just be a stability/motor control problem.

How to Perform the Stabilized Prone External Hip Rotation Test:

Instruct the patient to lie face down on the table/floor with the knee flexed. They will reach under the table with their arms and push upward into the table. This should activate the core and reposition the pelvis. Once stabilized, have the patient externally rotate their hip by allowing their lower leg to fall towards midline as far as possible. Repeat the test on the other side if needed.

Functional Criteria:

Normal range for external hip rotation is 40 degrees.

Additional Information:

Monitor the pelvis during stabilization to make sure a change occurs.

Possible Findings:

- FN: 40 degrees or greater of active external hip rotation, no pain
 - » Positive for Hip/Core stability/motor control dysfunction. Proceed to the Active External Tibial Rotation Test.
- FP: 40 degrees or greater of active external hip rotation, pain
 - » Proceed to the Passive Prone External Hip Rotation Test.
- DN: Less than 40 degrees of external hip rotation, no pain
 - » Proceed to the Passive Prone External Hip Rotation Test.
- DP: Less than 40 degrees of external hip rotation, pain
 - » Proceed to the Passive Prone External Hip Rotation Test.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Passive Prone Hip External Rotation Test:

To test passive hip external rotation in an unloaded posture with the hip extended to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone Hip External Rotation Test:

Instruct the patient to lie face down on the table/floor with the knee flexed. While stabilizing the pelvis, passively externally rotate their hip by lowering their lower leg towards midline, keeping the knee flexed.

Functional Criteria:

Normal range for external hip rotation is 40 degrees.

Additional Information:

Monitor for pelvic rotation.

Possible Findings:

- FN: 40 degrees or greater of passive external hip rotation, no pain
 - » Indicates an external hip rotation stability/motor control dysfunction. Proceed to the Active Seated External Tibial Rotation Test.
- FP: 40 degrees or greater of passive external hip rotation, pain
 - » Indicates pain in the hip with external hip rotation - complete local biomechanical exam. Proceed to the Active Seated External Tibial Rotation Test.
- DN: less than 40 degrees of passive external hip rotation, no pain
 - » Indicates a hip mobility dysfunction for external hip rotation - complete local biomechanical exam. Proceed to the Active Seated External Tibial Rotation Test.
- DP: less than 40 degrees of passive external hip rotation, pain
 - » Indicates a painful external hip rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Active Seated External Tibial Rotation Test.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Active Seated External Tibial Rotation Test:

To test active external tibial rotation in an unloaded lower-body posture.

How to Perform the Active Seated External Tibial Rotation Test:

Instruct the patient to get into a seated position with knees flexed to 90 degrees, sitting tall so the body is in an upright and erect posture, and arms down by their side. Have the person externally rotate the foot keeping the knee flexed. Repeat the test on the other side.

Functional Criteria:

Normal range for external tibial rotation is 20 degrees.

Additional Information:

When the knee is flexed to 90 degrees, a good picture of the total tibial rotation is presented. Monitor for excessive movement from the knees/hips and/or inversion/eversion.

Possible Findings:

- FN: 20 degrees or greater of active external tibial rotation, no pain
 - » Consider external tibial rotation normal. Proceed to the Lower Quarter Internal Rotation Flowchart.
- FP: 20 degrees or greater of active external tibial rotation, pain
 - » Proceed to the Passive Seated External Tibial Rotation Test.
- DN: Less than 20 degrees of active external tibial rotation, no pain
 - » Proceed to the Passive Seated External Tibial Rotation Test.
- DP: Less than 20 degrees of active external tibial rotation, pain
 - » Proceed to the Passive External Tibial Rotation Test.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Passive External Tibial Rotation Test:

To test passive external tibial rotation in an unloaded lower-body posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive External Tibial Rotation Test:

Instruct the patient to lay down in a supine position. Flex their knee and stabilize their femur. Passively externally rotate the tibia.

Functional Criteria:

Normal range for external tibial rotation is 20 degrees.

Additional Information:

When the knee is flexed to 90 degrees, a good picture of the total tibial rotation is presented.

Possible Findings:

- FN: 20 degrees or greater of passive external tibial rotation, no pain
 - » Indicates an external tibial rotation stability/motor control dysfunction. Proceed to the Lower Quarter Internal Rotation Flowchart.
- FP: 20 degrees or greater of passive external tibial rotation, pain
 - » Indicates pain with external tibial rotation - complete local biomechanical exam. Proceed to the Lower Quarter Internal Rotation Flowchart.
- DN: Less than 20 degrees of passive external tibial rotation, no pain
 - » Indicates an external tibial rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Lower Quarter Internal Rotation Flowchart.
- DP: Less than 20 degrees of passive external tibial rotation, pain
 - » Indicates an external tibial rotation mobility dysfunction with pain - complete local biomechanical exam. Proceed to the Lower Quarter Internal Rotation Flowchart.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Active Prone Internal Hip Rotation Test:

To test active hip internal rotation in an unloaded posture with the hip extended.

How to Perform the Active Prone Internal Hip Rotation Test:

Instruct the patient to lie face down on the table/floor with the knee flexed. Have the patient internally rotate their hip by allowing their lower leg to fall outwards toward you while keeping the knee flexed. Have the patient internally rotate the hip as far as possible. Repeat the test on the other side.

Functional Criteria:

Normal range for internal hip rotation is 30 degrees bilaterally.

Additional Information:

Monitor for pelvic rotation.

Possible Findings:

- FN: 30 degrees or greater of active internal hip rotation, no pain
 - » Consider internal hip rotation normal in the extended position. Proceed to the Active Seated Internal Tibial Rotation test.
- FP: 30 degrees or greater of active internal hip rotation, pain
 - » Proceed to the Stabilized Prone Internal Hip Rotation Test.
- DN: Less than 30 degrees of internal hip rotation, no pain
 - » Proceed to the Stabilized Prone Internal Hip Rotation Test.
- DP: Less than 30 degrees of internal hip rotation, pain
 - » Proceed to the Stabilized Prone Internal Hip Rotation Test.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Stabilized Prone Hip Internal Rotation Test:

To test if there is an underlying SMCD of the core that may be limiting hip mobility. Pelvic orientation can directly affect the spacing in the acetabulum and therefore give the impression that there is a hip joint mobility problem, when in reality it may just be a stability/motor control problem.

How to Perform the Stabilized Prone Hip Internal Rotation Test:

Instruct the patient to lie face down on the table/floor with the knee flexed. They will reach under the table with their arms and push upward into the table. This should activate the core and reposition the pelvis. Once stabilized, have the patient internally rotate their hip by allowing their lower leg to fall outwards toward you as far as possible. Repeat the test on the other side if needed.

Functional Criteria:

Normal range for internal hip rotation is 30 degrees.

Additional Information:

Monitor the pelvis during stabilization to make sure a change occurs.

Possible Findings:

- FN: 30 degrees or greater of active internal hip rotation, no pain
 - » Positive for Hip/Core stability/motor control dysfunction. Proceed to the Active Seated Internal Tibial Rotation Test.
- FP: 30 degrees or greater of active internal hip rotation, pain
 - » Proceed to the Passive Prone Internal Hip Rotation Test.
- DN: Less than 30 degrees of passive internal hip rotation, no pain
 - » Proceed to the Passive Prone Internal Hip Rotation Test.
- DP: Less than 30 degrees of passive internal hip rotation, pain
 - » Proceed to the Passive Prone Internal Hip Rotation Test.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Passive Prone Hip Internal Rotation Test:

To test passive hip internal rotation in an unloaded posture with the hip extended to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Prone Hip Internal Rotation Test:

Instruct the patient to lie face down on the table/floor with the knee flexed. While stabilizing the pelvis, passively internally rotate their hip by lowering their lower leg outwards toward you, keeping the knee flexed.

Functional Criteria:

Normal range for internal hip rotation is 30 degrees.

Additional Information:

Monitor for pelvic rotation.

Possible Findings:

- FN: 30 degrees or greater of passive internal hip rotation, no pain
 - » Indicates an internal hip rotation stability/motor control dysfunction. Proceed to the Active Seated Internal Tibial Rotation Test.
- FP: 30 degrees or greater of passive internal hip rotation, pain
 - » Indicates pain in the hip with hip internal rotation - complete local biomechanical exam. Proceed to the Active Seated Internal Tibial Rotation Test.
- DN: less than 30 degrees of internal hip rotation, no pain
 - » Indicates a hip mobility dysfunction for internal rotation - complete local biomechanical exam. Proceed to the Active Seated Internal Tibial Rotation Test.
- DP: less than 30 degrees of hip internal rotation, pain
 - » Indicates a painful internal hip rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Active Seated Internal Tibial Rotation Test.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Active Seated Internal Tibial Rotation Test:

To test active internal tibial rotation in an unloaded lower-body posture.

How to Perform the Active Seated Internal Tibial Rotation Test:

Instruct the patient to get into a seated position with knees flexed to 90 degrees, sitting tall so the body is in an upright and erect posture, and arms down by their side. Have the person internally rotate the foot keeping the knee flexed. Repeat the test on the other side.

Functional Criteria:

Normal range for internal tibial rotation is 20 degrees.

Additional Information:

When the knee is flexed to 90 degrees, a good picture of the total tibial rotation is presented. Monitor for excessive movement from the knees/hips and/or inversion/eversion.

Possible Findings:

- FN: 20 degrees or greater of active internal tibial rotation, no pain
 - » Consider internal tibial rotation normal.
- FP: 20 degrees or greater of active internal tibial rotation, pain
 - » Proceed to the Passive Internal Tibial Rotation Test.
- DN: Less than 20 degrees of internal tibial rotation, no pain
 - » Proceed to the Passive Internal Tibial Rotation Test.
- DP: Less than 20 degrees of internal tibial rotation, pain
 - » Proceed to the Passive Internal Tibial Rotation Test.



MULTI-SEGMENTAL ROTATION BREAKOUTS



Test Objective for the Passive Internal Tibial Rotation Test:

To test passive internal tibial rotation in an unloaded lower-body posture to differentiate between mobility and stability/motor control dysfunction.

How to Perform the Passive Internal Tibial Rotation Test:

Instruct the patient to lie down in a supine position. Flex their knee and stabilize their femur. Passively internally rotate the tibia.

Functional Criteria:

Normal range for internal tibial rotation is 20 degrees.

Additional Information:

When the knee is flexed to 90 degrees, a good picture of the total tibial rotation is presented.

Possible Findings:

- FN: 20 degrees or greater of passive internal tibial rotation, no pain
 - » Indicates an internal tibial rotation stability/motor control dysfunction.
- FP: 20 degrees or greater of passive internal tibial rotation, pain
 - » Indicates pain with internal tibial rotation - complete local biomechanical exam.
- DN: Less than 20 degrees of passive internal tibial rotation, no pain
 - » Indicates an internal tibial rotation mobility dysfunction - complete local biomechanical exam.
- DP: Less than 20 degrees of passive tibial internal rotation, pain
 - » Indicates a painful internal tibial rotation mobility dysfunction - complete local biomechanical exam.

